

## Unit 102 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 36.3 ac. 18 plots, BAF 20, variable-radius cruise on a 300 ft grid, 197 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

**TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE**

| STAND DENSITY        |                                        | STAND INFO          |                                                                                                                                       |
|----------------------|----------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Basal area           | 219 sq ft/ac (1 SE of mean, BA 2.7 %)  | Type                | Sierran mixed conifer                                                                                                                 |
| Trees per acre       | 88                                     | CWHR size / density | 4 / D (cover 87 %)                                                                                                                    |
| QMD                  | 21.3 in                                | SDI                 | 285 (50 % of max 574)                                                                                                                 |
| SAWTIMBER (>= 10 in) |                                        | BIOMASS (< 10 in)   |                                                                                                                                       |
| BA   QMD             | 218 sq ft/ac   21.5 in                 | BA   QMD            | 1 sq ft/ac   9.8 in                                                                                                                   |
| TPA                  | 86                                     | TPA                 | 2                                                                                                                                     |
| Volume               | 34.1 MBF/ac (6796 cu ft)               | Volume              | 0.3 green tons/ac (15 cu ft)                                                                                                          |
| TREATMENT TARGET     |                                        | CRUISE DESIGN       |                                                                                                                                       |
| Leave                | 35 % of max SDI, about 155 sq ft/ac BA | Vol. sampling error | 12.9 % at 95 % (CV 26 %, t = 2.11, n = 18)                                                                                            |
| Species (BA)         | PP 69, WF 66, DF 33, IC 28, SP 23      | Stratum standard    | 30 % of volume (R5 2409.12-2021-4, sec. 41.1): <b>MEETS</b> ; plots for standard 4                                                    |
|                      |                                        | Sale volume share   | 1.6 % of sale volume: standard shown for information (under the R5 10 % threshold)                                                    |
|                      |                                        | BA sampling error   | 5.7 % at 95 % (CV 12 %); reference, not the standard                                                                                  |
|                      |                                        | Measured trees      | PP 62, WF 59, DF 30, SP 21, IC 25; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): <b>ok</b> |
|                      |                                        | Plot count          | 18; R5 20-plot minimum for an area-based cruise (sec. 41.3): <b>SHORT</b> by 2                                                        |

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise\_data.xlsx, Standards and assumptions.

**TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE**

| DBH class, in | Trees      | TPA         | BA           | cu ft       | MBF         |
|---------------|------------|-------------|--------------|-------------|-------------|
| 8-11.9        | 6          | 10.7        | 6.7          | 124         | 0.6         |
| 12-15.9       | 18         | 19.6        | 20.0         | 394         | 2.0         |
| 16-19.9       | 32         | 20.2        | 35.6         | 897         | 4.5         |
| 20-23.9       | 40         | 17.1        | 44.4         | 1299        | 6.5         |
| 24-27.9       | 22         | 6.7         | 24.4         | 770         | 3.9         |
| 28-31.9       | 30         | 7.1         | 33.3         | 1104        | 5.5         |
| 32-35.9       | 19         | 3.4         | 21.1         | 744         | 3.7         |
| 36-39.9       | 8          | 1.1         | 8.9          | 365         | 1.8         |
| 40-43.9       | 10         | 1.2         | 11.1         | 441         | 2.2         |
| 44-47.9       | 5          | 0.5         | 5.6          | 240         | 1.2         |
| 48-51.9       | 1          | 0.1         | 1.1          | 66          | 0.3         |
| 52-55.9       | 3          | 0.2         | 3.3          | 180         | 0.9         |
| 56-59.9       | 1          | 0.1         | 1.1          | 73          | 0.4         |
| 60-63.9       | 2          | 0.1         | 2.2          | 115         | 0.6         |
| <b>All</b>    | <b>197</b> | <b>88.1</b> | <b>218.9</b> | <b>6811</b> | <b>34.2</b> |

**TABLE 3. STOCK TABLE BY SPECIES, PER ACRE**

| Species    | Trees      | TPA         | BA           | QMD         | cu ft       | MBF         |
|------------|------------|-------------|--------------|-------------|-------------|-------------|
| PP         | 62         | 28.1        | 68.9         | 21.2        | 2266        | 11.4        |
| WF         | 59         | 26.4        | 65.6         | 21.3        | 2042        | 10.2        |
| DF         | 30         | 15.3        | 33.3         | 20.0        | 1016        | 5.1         |
| IC         | 25         | 9.1         | 27.8         | 23.6        | 755         | 3.8         |
| SP         | 21         | 9.2         | 23.3         | 21.6        | 733         | 3.7         |
| <b>All</b> | <b>197</b> | <b>88.1</b> | <b>218.9</b> | <b>21.3</b> | <b>6811</b> | <b>34.2</b> |

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 102 plots, 300 ft grid (orange: QA flag)

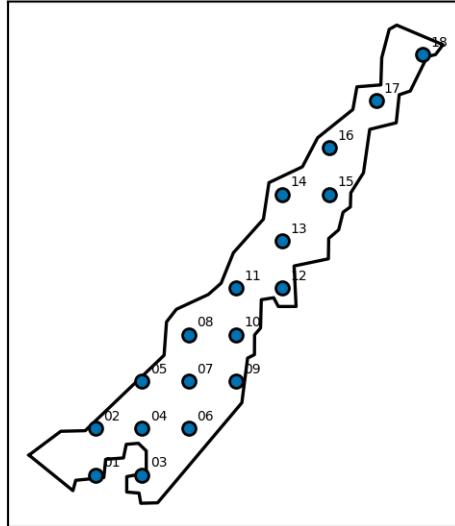


Figure 1. Plot locations with QA status.

**QA findings:** none; unit accepted; 2 more plots to reach the R5 20-plot minimum (sec. 41.3)

TABLE 4. PLOT LIST

| Plot   | Trees in | BA sq ft/ac | Slope % | Status |
|--------|----------|-------------|---------|--------|
| 102-01 | 12       | 240         | 4       | ok     |
| 102-02 | 11       | 220         | 19      | ok     |
| 102-03 | 9        | 180         | 5       | ok     |
| 102-04 | 11       | 220         | 5       | ok     |
| 102-05 | 10       | 200         | 15      | ok     |
| 102-06 | 12       | 240         | 4       | ok     |
| 102-07 | 8        | 160         | 4       | ok     |
| 102-08 | 12       | 240         | 27      | ok     |
| 102-09 | 10       | 200         | 5       | ok     |
| 102-10 | 11       | 220         | 8       | ok     |
| 102-11 | 12       | 240         | 19      | ok     |
| 102-12 | 12       | 240         | 9       | ok     |
| 102-13 | 10       | 200         | 6       | ok     |
| 102-14 | 11       | 220         | 19      | ok     |
| 102-15 | 10       | 200         | 4       | ok     |
| 102-16 | 12       | 240         | 22      | ok     |
| 102-17 | 11       | 220         | 19      | ok     |
| 102-18 | 13       | 260         | 36      | ok     |